CS 791: Quiz 1

1. Given a square matrix
$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

   (a) Find the eigenvalues of A

      i. 3, 4

      ii. 1, 10

      iii. 2, 7

      iv. 5, 2

   ..1

   Solve for λ in $\det(A - I\lambda) = \begin{vmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} = 0$ to get the eigenvalues. $\lambda = 5, 2$

   (b) Do the eigen decomposition in the form $A = PDP^{-1}$, $D = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ where $a > b$, $a, b \in \mathbb{N}$, $\gcd(a,b) = 1$, $P = \begin{bmatrix} x & -1 \\ 1 & y \end{bmatrix}$. Find $\mathrm{Tr}(P)$. ..1

   Here, a and b are the eigenvalues calculated above. Let $\mathbf{v}_1 = (v_1^1, v_1^2)$ and $\mathbf{v}_2 = (v_2^1, v_2^2)$ the eigenvectors calculated using $A\mathbf{v} = \lambda\mathbf{v}$ for eigenvalues a and b respectively and P will be of the form $\begin{bmatrix} v_1^1 & v_2^1 \\ v_1^2 & v_2^2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$. So, $\mathrm{Tr}(P) = 2 + 1 = 3$

2. Given a joint pdf $f_{X,Y}(x,y) = kxy$ where $(x,y) \in (0,1)$. Find the conditional pdf $f_{X|Y}(x|y)$ for $x = 0.5$ and $y = 0.8$. ..1

   $f_Y(y) = \int_0^1 f_{X,Y}(x,y)dx = \frac{ky}{2}$ and $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = 2x = 1$

3. If the pdf of Z is $f_Z(z) = \frac{k}{z^2+1}$, let the pdf of $X = Z^2$ be $g_X(x)$. Find $\frac{g_X(4)}{g_X(9)}$. ..2

   Let $G_X(x)$ and $F_Z(z)$ be the cdf of $g_X(x)$ and $f_Z(z)$ respectively.

$$G_X(x) = P(Z^2 \leq x) = P(-\sqrt{x} \leq Z \leq \sqrt{x})$$
$$= F_Z(\sqrt{x}) - F_Z(-\sqrt{x})$$
$$g_X(x) = \frac{1}{2\sqrt{x}} f_Z(\sqrt{x}) + \frac{1}{2\sqrt{x}} f_Z(-\sqrt{x})$$

   On substituting the values, $g_X(4) = \frac{k}{10}$ and $g_X(9) = \frac{k}{30}$. Thus $\frac{g_X(4)}{g_X(9)} = 3$.

4. $X_1, X_2, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables from an exponential distribution with parameters $\lambda_1, \lambda_2, \ldots, \lambda_n$ respectively. Given the cdf of exponential distribution:
$$F_{X_i}(x) = 1 - e^{-\lambda_i x}, \quad \lambda_i = i \cdot 2^i$$
   The cdf of $\min(X_1, X_2, \ldots, X_n)$ is in the form $1 - e^{-ax}$. Find a.

   (a) $(n-1)2^{n+1} + 2$

(b) $(n-1) \cdot 2^{n+1}$

(c) $(n-1)2^n + 4n - 2$

(d) $2^{n+1} + 2$

..1

$F_{X_i}(x) = P(X_i < x) = 1 - e^{-\lambda_i x}$. So, $P(X_i > x) = e^{-\lambda_i x}$

The cdf of $\min(X_1, X_2, \ldots, X_n)$ is

$P(\min(X_1, X_2, \ldots, X_n) < x) = 1 - P(\min(X_1, X_2, \ldots, X_n) > x)$

$$P(\min(X_1, X_2, \ldots, X_n) > x) = P(X_1 > x, X_2 > x, \ldots, X_n > x)$$

$$= \prod_{i=1}^{n} P(X_i > x) \text{ (due to independence)}$$

$$= \prod_{i=1}^{n} e^{-\lambda_i x}$$

$$= e^{-x \sum_{i=1}^{n} \lambda_i}$$

Therefore, $a = \sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} i \cdot 2^i = (n-1)2^{n+1} + 2$.

5. A factory has two machines, A and B, producing items. Machine A produces 60% of the items, and Machine B produces 40%. The probability of a defective item from Machine A is 5%, and from Machine B is 2%. Find the probability that a defective item is made by Machine B. Let $P(\text{defective from B}) = p/q$ with $\gcd(p, q) = 1$. Give $q - 2p$ as the answer. ..1

According to Bayes' rule,

$$P(B \mid defective) = \frac{P(B)P(defective \mid B)}{P(A)P(defective \mid A) + P(B)P(defective \mid B)}$$

On substituting the values we get $\frac{4}{19}$.

6. Let X be from an exponential distribution with pdf $\lambda e^{-\lambda x}$. Observations: $1, 3, 1, 3, 2$. Find the maximum likelihood estimate (MLE) of λ. ..1

Let the observations be $x_1, x_2, \cdots, x_n$. We have to calculate

$$\text{argmax}_\lambda \prod_{i=1}^{n} p(x_i \mid \lambda)$$

$$= \text{argmax}_\lambda (\lambda^n e^{-\lambda \sum_{i=1}^{n} x_i})$$

$$= \text{argmax}_\lambda (n \log \lambda - \lambda \sum_{i=1}^{n} x_i)$$

Taking logarithm does not affect argmax.

On taking the derivative w.r.t. λ, we get

$$\frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0$$

From this, we get $\lambda = \frac{n}{\sum_{i=1}^{n} x_i}$. On substituting the values, we get $\lambda = 0.5$.

7. If regularisation parameter increases, what happens to bias and variance? ..1

   (a) Bias increases, Variance increases
   (b) Bias decreases, Variance decreases
   (c) Bias increases, Variance decreases
   (d) Bias decreases, Variance increases

   Refer section 5.4.4 in chapter 5 of the prerequisite reading material. As the regularization parameter increases, overfitting decreases. The correct answer is (c).

8. Let

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 9 \\ 7 \end{bmatrix}, \quad x = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

   If $Ax = b$, then find pqr. ..1

   Basic linear algebra. On solving, x = $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. So, the answer is 4.

9. Find the Frobenius norm of

$$\begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$$

..1

   Refer section 2.5 in chapter 2 of the prerequisite reading material.

$$\|A\|_F = \sqrt{\sum_{i,j} A_{i,j}^2}$$

$$= \sqrt{1^2 + 2^2 + 4^2 + 2^2} = 5$$

10. Which of these functions is numerically unstable for large $x > 0$? ..1

   (a) $\log(x)$
   (b) $\exp(-x)$
   (c) sigmoid(x)
   (d) $\exp(x)$

   $\exp(x)$ grows exponentially and doesn't stabilize like the others for large x

11. Let $f(x) = (x-4)^2$, starting from $x = 0$:

(a) Compute x After 1 step of gradient descent with learning rate $\alpha = 0.1$. ..1

1 step of gradient descent is - $x_{new} = x_{old} - \eta \frac{df(x)}{dx}$

$x_{new} = x - 0.1 \cdot 2(x-4) \implies x_{new} = 0.8$ (Here $x = 0$)

(b) Compute x after After 2 steps of gradient descent with learning rate $\alpha = 0.1$. ..1

$x_{new} = x - 0.1 \cdot 2(x-4) \implies x_{new} = 1.44$ (Here $x = 0.8$)

12. Let $x \in \mathbb{R}^n$ be the input feature vector, $w \in \mathbb{R}^n$ the weight vector, and $b \in \mathbb{R}$ the bias. In logistic regression, the predicted probability is given by:

$$\hat{y} = \frac{1}{1+e^{-z}}, \quad \text{where} \quad z = w^T x + b$$

The binary cross-entropy loss for a single training example with label $y \in \{0, 1\}$ is:

$$L(w, b) = -\left[y \log(\hat{y}) + (1-y)\log(1-\hat{y})\right]$$

(a) Derive an expression for the gradient $\frac{dL}{dw}$

i. $(\hat{y} - y)x$
ii. $(y - \hat{y})x$
iii. $(y - \hat{y})$
iv. $y\hat{y}x$

..2

$\frac{dL}{dw} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{dw} = -[\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}][\frac{e^{-z}}{(1+e^{-z})^2}][x] = -[\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}][\hat{y}(1-\hat{y})][x] = (\hat{y} - y)x$

(b) Derive an expression for the gradient $\frac{dL}{db}$

i. $(y - \hat{y})x$
ii. $(\hat{y} - y)$
iii. $(y - \hat{y})$
iv. $y\hat{y}$

..1

$\frac{dL}{db} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{db} = -[\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}][\frac{e^{-z}}{(1+e^{-z})^2}][1] = -[\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}][\hat{y}(1-\hat{y})] = (\hat{y} - y)$

Total: 17